COVID Mortality Disruption Lab

Research Question: Did COVID-19 disrupt the trend for six other leading causes of death, has it recovered, and where did it hit hardest?

By: Aidan Chi

Summary

I tested six causes of death against their own pre-pandemic trends using CDC WONDER mortality data from 1999 to 2024. **Five of them moved significantly off trend in 2020-21**, and five of those results hold up after correcting for testing six causes at once. Much of that has since faded. By 2024, three causes were significantly off trend: diabetes (+15.0%) and cancer (+4.6%) above it, and Alzheimer's disease (-19.1%) below it. Heart disease, stroke, and drug overdose were back within their normal range. The 2024 results for heart disease, diabetes, stroke, and drug overdose depend on which baseline window is used (section 4).

"Trend" means where each cause's pre-2020 trend line pointed, not its 2019 level. Diabetes's 2024 rate was back at its 2019 level (21.7 vs. 21.6); it reads as above trend because its trend line was still falling. Stroke is back within its normal range but still 4% above its 2019 rate. Drug overdose is back within its normal range but still 7% above its 2019 rate. Cancer kept falling (146.2 in 2019, 139.4 in 2024), just more slowly than its trend.

"Disrupted" here means statistically distinguishable from a cause's normal year-to-year variation before 2020, not simply far from zero: the gap falls outside the range that variation would explain. That range is narrow for steady causes and wide for volatile ones: about ±1% for cancer in 2024, but about ±13% for drug overdose. So cancer's +4.6% counts, while drug overdose's -4.3% doesn't. Being within the normal range doesn't mean no effect, only that the gap can't be told apart from ordinary year-to-year swings with this method.

Two results went against what I predicted before looking at the data. Cancer was expected to show nothing in this time window and showed a small increase, though that result depends on the shape of the trend. Alzheimer's disease was expected to show a large increase and showed none in 2020-21, though a decline appears in later years. A negative control, four sensitivity checks, and a multiple-testing correction were all run before any result was reported.

COVID-19 itself is not what's being studied here. It's treated as a shock to the health system, and the question is what happened to deaths from other causes afterward. Mortality data can show what changed,

by how much, and where. It can't show why, because direct infection, delayed care, isolation, and economic stress all leave the same mark in the numbers. Every result below is an association, not a causal claim.

This kind of excess-mortality analysis is how public health agencies estimate a pandemic's toll beyond deaths officially attributed to the virus: missed cancer screenings, interrupted addiction treatment, delayed heart care. Each cause got a written prediction and confidence level before any 2020-2024 data was pulled, so the results can be checked against what was expected rather than explained after the fact.

1. Research question

> Which causes of death were most disrupted during the COVID-19 pandemic (2020-2024), how long did those disruptions last, and how did their size vary across U.S. counties by income, health insurance coverage, and rurality?

The six causes tested are heart disease, diabetes, Alzheimer's disease, stroke, drug overdose, and cancer. Two more series fill out the design: congenital malformations as a negative control (a cause with no direct link to COVID), and COVID-19 deaths as a reference that isn't itself tested.

2. Methods

Excess mortality with a fixed breakpoint. For each cause, I fit a straight-line trend to its pre-pandemic age-adjusted death rate and projected it forward through 2020-2024 with a 95% prediction interval. A year counts as disrupted when the observed rate falls outside that interval. Most causes use a 1999-2019 baseline; heart disease and stroke use 2010-2019 for reasons covered in section 4. The breakpoint is 2020 because that's when the pandemic started, not because a search found it.

How long it lasted. Causes with a significant 2020-21 disruption are sorted into three groups based on the years 2022-2024. *Persisted* means the later years that were still significantly off trend leaned the same way as 2020-21. It doesn't require the most recent year to be off trend. *Resolved* means no later year was significantly off trend, and *Reversed* means the later significant years leaned the opposite way.

Cross-check. Three change-point methods (PELT, binary segmentation, and segmented regression) were run on each series to see whether they find a break near 2020 on their own. They search the whole series for the single best breakpoint, so landing somewhere else doesn't count against a result at the fixed 2020 date.

Negative control. The same pipeline was run on a cause with no direct COVID link. If it had shown a significant disruption, the method would be picking up an artifact, and no other result could be trusted.

Multiple testing. Testing six causes at once makes a chance "significant" result more likely, so the six p-values are adjusted together with the Benjamini-Hochberg false discovery rate (FDR) method. The county analysis gets its own correction for each cause.

Full-period check. Each cause also gets a second p-value that pools all five post-2020 years instead of just 2020-21. It doesn't replace the main test, since averaging more years can hide a reversal as easily as it can reveal a slow effect, but it catches disruptions that showed up after 2021 (see Alzheimer's in section 3).

Autocorrelation check. The main test assumes each baseline year's deviation from trend is independent of the last. For several causes it isn't, so the same test is also run with Newey-West (HAC) standard errors, which allow for that (section 4).

Data. CDC WONDER Underlying Cause of Death, from two databases joined at their 2018-2019 overlap: D76 (1999-2020) for the baseline and D158 (2018-2024, Single Race) for 2020-2024. County data for the five disrupted causes compares pooled 2015-2019 with pooled 2020-2024 rates and is matched to County Health Rankings & Roadmaps measures.

3. Results

Figure 1 shows the method on drug overdose, the cause with the largest jump. The solid line is the observed death rate. The dashed gray line is where the 1999-2019 trend says the rate should have gone, and the two thin pink lines mark its 95% prediction interval. Overdose deaths broke above that interval in 2020 and stayed there through 2023, then fell back in 2024.

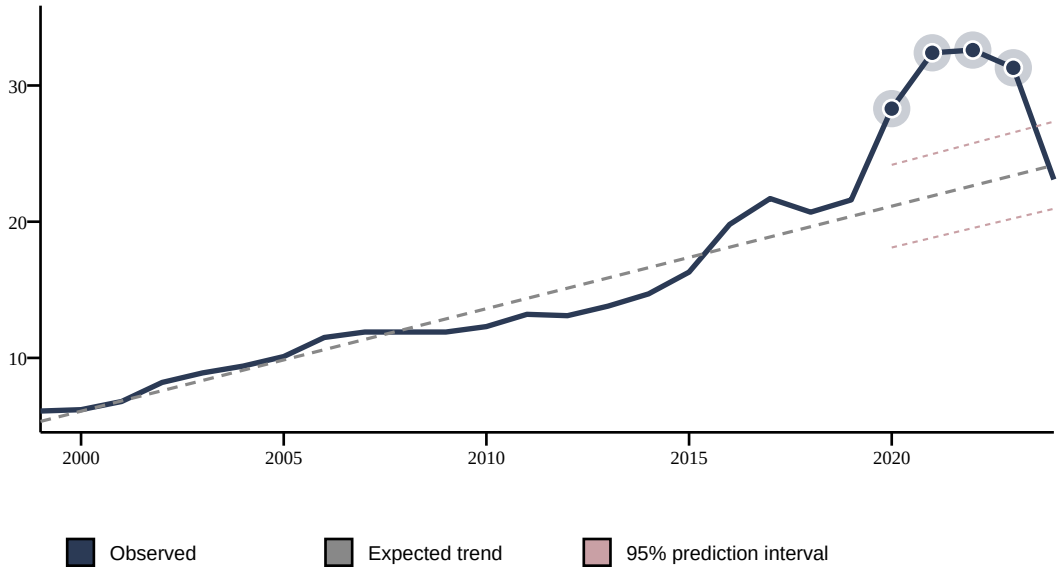

Figure 1. Drug overdose: observed death rate against its pre-pandemic trend, 1999-2024.

5 of 6 causes show a significant 2020-21 disruption, and **5 of 6** hold up after FDR correction.

Cause	Result	p-value	FDR-sig.	2020-21 dev.	2024 dev.	Robust?
Drug overdose	Persisted	2.07e-07	Yes	+40.9%	-4.3%	Yes
Diabetes	Persisted	2.51e-06	Yes	+26.9%	+15.0%	Yes
Heart disease	Persisted	0.000112	Yes	+7.8%	+3.0%	Yes
Cancer	Persisted	0.000446	Yes	+1.7%	+4.6%	No
Stroke	Persisted	0.00403	Yes	+8.8%	+6.1%	No
Alzheimer's disease	No significant disruption	0.822	No	+1.0%	-19.1%	Yes

"Dev." is the effect size: how far the observed rate sits from the expected trend, in percent. A result can be significant and still small. "Robust?" shows whether the result survives a curved or steady-percentage trend in place of a straight line; stroke fails the curved trend and cancer the steady-percentage one (section 4).

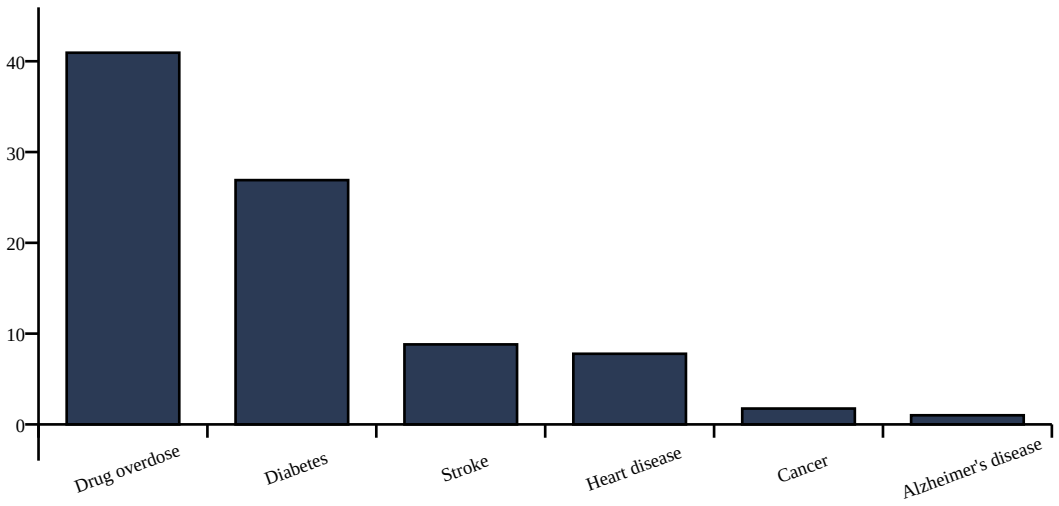

Figure 2. Average 2020-21 deviation from the expected trend, by cause.

Figure 3 repeats Figure 1 for all six causes, with each cause's p-value in its panel title. It's worth noticing how poorly the size of a gap predicts significance. Cancer's gap is barely visible but significant, because its year-to-year noise before 2020 was so small. The next two sections explain why some results hold up and others need caution.

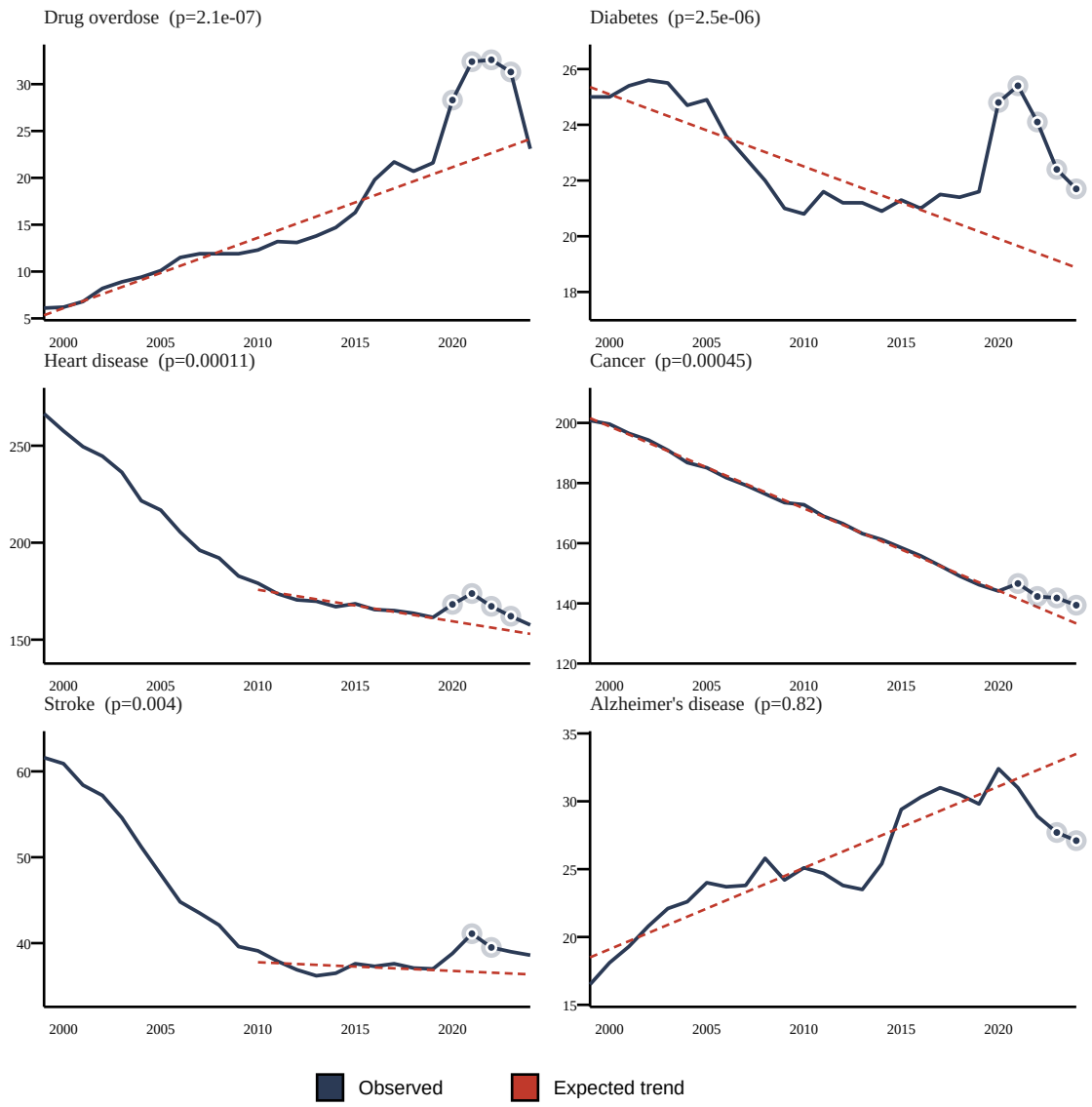

Figure 3. All six causes, observed vs. expected trend, 1999-2024.

Two results that went against the predictions

Cancer was expected to show nothing. The prediction was a null result, since missed screenings were thought to take longer than this data window to show up as deaths. Instead cancer shows a significant disruption (p = 0.000446, survives FDR correction), though a small one: +1.7% in 2020-21 and +4.6% in 2024. It's also fragile: with a trend that falls by a steady percentage each year, the 2020-21 gap is no longer significant. Other research doesn't settle this either. Early models predicted large increases in cancer deaths from delayed screening, while later work using English data found lung and breast cancer deaths came in below pre-pandemic trends.

Alzheimer's was expected to show a large effect. The reasoning was that isolation and disruption in care facilities would show up clearly in dementia deaths. This project's annual test didn't find it (p = 0.82 for 2020-21). That doesn't rule out an effect: 2020 alone ran +4% above trend before 2021 fell below it, and studies that also count dementia as a contributing cause found large excess deaths in 2020. Pooling all five post-2020 years does find a later decline (p = 0.016, averaging -8.7% vs. trend). One possible explanation is mortality displacement: people who would have died in 2023-2024 may have died earlier in the pandemic.

The main classification still uses the pre-registered 2020-21 window, so this is reported as an additional finding.

Negative control

Congenital malformations, a cause concentrated in infancy with no direct COVID link, shows no significant disruption, as expected (p = 0.706 on raw death counts). Its age-adjusted *rate* does drift +14% above trend by 2024, which is why the gate uses raw death counts: WONDER reports this rate to one decimal place at about 3 per 100,000, and the rate-based test is too sensitive to that rounding. Passing doesn't prove every other result is real, but failing would have been strong evidence of a problem.

The first negative control, accidental drowning, failed. Drowning deaths really did rise from 2020 on, in line with CDC reports tied to pool and beach closures and lifeguard shortages. It was replaced because it turned out to be affected by the pandemic, not because the method broke. The replacement isn't perfectly insulated either, since prenatal and obstetric care were also disrupted in 2020-21, but that path is smaller and less direct than for the six test causes.

4. Robustness

A baseline correction for heart disease and stroke. Both originally used the same 1999-2019 baseline as the other causes, and both failed the trend-shape check below. Using only 1999-2019 data, an F-test comparing straight-line and curved fits showed strong curvature for these two causes (F=222.7 and F=162.6, both p<0.00001) and not for the others (F<9.5). Both fell steeply through the 2000s and then leveled off, so a straight line across the whole period already sat 17.7 and 5.8 points below the real 2019 rate before the pandemic. A curved fit matches history well but predicts *rising* rates after 2019, a common problem when extending a polynomial past its data, so it wasn't used. A straight line over 2010-2019 has neither problem and is now the baseline for these two causes. Both stay significant, but their estimated 2020-21 deviation drops from 26-37% to 8-9%. Details are in research_protocol.md (2026-09-01 addendum).

With that correction in place, four sensitivity checks re-run the main test with one modeling choice changed at a time.

Baseline window (1999-2019 vs. 2010-2019). No result changes.

Significance threshold (α=0.05 vs. α=0.01). One cause changes: Stroke.

Baseline shape (straight vs. curved). One cause changes: Stroke.

Trend scale (steady amount vs. steady percentage per year). One cause changes: Cancer.

The trend-shape checks matter most. Heart disease passes both. Stroke doesn't survive a curved baseline (p = 0.18) and also changes at the stricter significance level, which makes it the least certain "Persisted" result in this report. Cancer's small 2020-21 gap isn't significant with a steady-percentage trend (p = 0.43), though cancer is still above trend in 2024 under it. Diabetes and drug overdose pass every check.

The 2024 status depends more on the baseline window. The checks above test the 2020-21 result, which holds under either window for every cause. Each cause's 2024 status was also recomputed under the other window. Primary results stay on the pre-registered windows (research_protocol.md, 2026-10-19 reversal addendum); for heart disease and stroke the other window is the 1999-2019 line already shown to misfit them.

Cause	Primary window	2024 gap	Other window	2024 gap

Heart disease	2010-2019	+3.0% (within)	1999-2019	+34.9% (above)
Diabetes	1999-2019	+15.0% (above)	2010-2019	+0.2% (within)
Alzheimer's disease	1999-2019	-19.1% (below)	2010-2019	-23.9% (below)
Stroke	2010-2019	+6.1% (within)	1999-2019	+57.3% (above)
Drug overdose	1999-2019	-4.3% (within)	2010-2019	-18.0% (below)
Cancer	1999-2019	+4.6% (above)	2010-2019	+5.4% (above)

Lag-1 autocorrelation measures whether a year that runs above trend tends to be followed by another one. In the pre-pandemic baselines it is high for diabetes, overdose, and Alzheimer's (0.65-0.82), moderate for stroke (0.50), and low for heart disease and cancer (0.12-0.19). Heart disease and stroke had much higher values (0.92 and 0.93) on the old 1999-2019 baseline, which is another sign that a straight line was the wrong fit there. Because the standard test assumes no autocorrelation, the same test was rerun with Newey-West (HAC) standard errors. That raises the p-values most for diabetes (about 25 times) and drug overdose (about 120 times), but all 5 significant causes stay significant.

Cause	Autocorrelation	Classical p-value	HAC p-value
Drug overdose	0.75	2.07e-07	2.53e-05
Heart disease	0.12	0.000112	2.7e-05
Diabetes	0.82	2.51e-06	6.66e-05
Cancer	0.19	0.000446	0.000659
Stroke	0.50	0.00403	0.00205
Alzheimer's disease	0.65	0.822	0.857

Joining the two databases. In the 2018-2019 years both databases cover, every cause matched within the 10% reliability threshold, and the median difference was 0% for every cause.

5. Which counties were hit hardest

Each of the five causes with a 2020-21 disruption has county data. For each county, disruption is the change in death rate from 2015-2019 to 2020-2024, using one five-year total per period, and it's compared against five County Health Rankings & Roadmaps measures. A county is included when CDC reports a reliable rate in both periods, which takes roughly 20 deaths over five years. County rates are crude rather than age-adjusted, because WONDER doesn't offer age adjustment at the county level for its 2018-2024 database, so some of a county's change could come from its population getting older.

Cause	Counties included	Median change (per 100,000)
Heart disease	3,066	+15.4
Diabetes	2,342	+4.1
Stroke	2,651	+2.7
Drug overdose	1,564	+7.5
Cancer	3,062	+2.3

What went with larger or smaller disruption (FDR-significant only):

• For all five causes, higher smoking rates went with larger disruption.

• For all five causes, higher uninsured rates went with larger disruption.

• Higher obesity rates went with larger disruption for heart disease, diabetes, and drug overdose; no significant link for stroke and cancer.

• Higher household income went with smaller disruption for heart disease, diabetes, drug overdose, and cancer; no significant link for stroke.

• Higher rurality went with larger disruption for heart disease, drug overdose, and cancer; smaller disruption for stroke; no significant link for diabetes.

County measure	Heart	Diabetes	Stroke	Overdose	Cancer
% uninsured	+77.9*	+15.2*	+15.3*	+16.8*	+35.5*
% adult smokers	+199.5*	+49.7*	+26.9*	+84.6*	+40.2*
% adult obesity	+130.0*	+35.6*	+8.9	+40.2*	+15.6
Income (per $10k)	-4.1*	-1.2*	-0.3	-2.1*	-0.8*
% rural	+13.6*	+1.0	-3.2*	+2.3*	+7.3*

Regression slopes: change in death rate per 100,000 for each unit of the measure (a full 0-1 share for the percentage measures; $10,000 for income). * = significant after FDR correction within that cause.

The rurality result changed

An earlier version of this analysis needed a reliable rate in at least two individual years of each period. That kept only 926 counties for diabetes and 641 for drug overdose, mostly larger and less rural ones, and it pointed to smaller disruption in more rural counties (slope -3.6, significant, for diabetes; -3.9, p = 0.07, for drug overdose). With five-year totals, which keep most rural counties, that pattern disappears for diabetes and reverses for drug overdose. The switch was made after the earlier results were known, so it's logged in research_protocol.md (2026-10-19 addendum) and both versions are shown in the app.

Two caveats still apply. Counties left out are still mostly rural (83-94% rural, vs. 45-63% for included counties), though far fewer are left out than before. And rural counties tend to be older and aging faster, which crude rates don't adjust for. Splitting the included counties at their median rurality:

• **Heart disease:** more rural half -1.7 (p=0.88), less rural half +23.2 (p=4.1e-09). The overall result comes mainly from the less rural half; the other half shows no significant link.

• **Diabetes:** more rural half -5.4 (p=0.041), less rural half +4.7 (p=0.0074). The two halves point in opposite directions, so the overall rurality result isn't one simple pattern.

• **Stroke:** more rural half -5.7 (p=0.053), less rural half -5.1 (p=0.0048). The overall result comes mainly from the less rural half; the other half shows no significant link.

• **Drug overdose:** more rural half +5.5 (p=0.043), less rural half -2.7 (p=0.3). The overall result comes mainly from the more rural half; the other half shows no significant link.

• **Cancer:** more rural half +1.7 (p=0.84), less rural half +10.6 (p=3.8e-05). The overall result comes mainly from the less rural half; the other half shows no significant link.

Exploratory: Alzheimer's disease. It had no 2020-21 disruption, so it isn't part of the county analysis above, but it was run the same way for comparison (2,426 counties). None of the five county measures survives FDR correction for it. Its national decline came in 2023-24, and averaging all of 2020-24 mixes that with earlier years, so this is a blunt measure of it.

This analysis has no data on local policies or on age and race, so it doesn't try to explain any single county, and these are associations, not causes.

6. Limitations

• This is an observational study of population averages, so it can't support causal claims about individuals. At the county level there's also a risk of ecological fallacy: a pattern across counties may not hold for the people living in them.

• Counties with too few deaths for a reliable rate are left out, and they're mostly rural. Pooling five years keeps far more of them than the earlier method did, but the most rural places are still under-represented, and the rurality result differs by cause (section 5).

• The two CDC WONDER databases may not line up perfectly. The 2018-2019 overlap check reduces this risk but can't remove it.

• Death certificate coding may have shifted in 2020-21 while reporting systems were strained, separate from any real change in deaths.

• Cancer was predicted to show no disruption and showed a small one. The cause isn't clear, and its +4.6% deviation in 2024 is modest.

• The standard test assumes baseline years are independent, which isn't true for several causes (autocorrelation 0.65-0.82 for diabetes, overdose, and Alzheimer's; 0.50 for stroke; 0.12-0.19 for heart disease and cancer). The HAC version of the test accounts for this, and all 5 significant causes stay significant under it (section 4).

• Significant doesn't mean large. Drug overdose and diabetes had the biggest 2020-21 jumps (roughly 27-41%), while cancer, heart disease, and stroke were significant but much smaller (about 2-9%).

• Heart disease and stroke use a 2010-2019 baseline instead of 1999-2019 because the longer window didn't fit their pre-pandemic trend (section 4). Heart disease now holds up under either baseline shape. Stroke and cancer are the least certain results: each loses significance under one alternative trend shape.

• County rates are crude, not age-adjusted, so some measured disruption may reflect each county's population getting older.

• Small counties are suppressed or unstable in the county data.

• Smoking and obesity rates come from modeled small-area estimates (PLACES), not direct counts.

• The county measures come from the 2024 County Health Rankings release, so they describe counties during or after the pandemic, not before it. Income and insurance rates may have changed during the pandemic itself.

• Many tests were run: across the six causes, and across the county measures for each cause. Both sets are corrected separately.

• Neighboring counties aren't independent of each other, and this analysis doesn't model that spatial correlation.

• Mortality data alone can't separate direct infection, delayed care, isolation, and economic stress. Every finding here is an association.

7. Data and reproducibility

National results use 15 CDC WONDER exports (7 from the 1999-2020 database, 8 from the 2018-2024 database). County results use 10 pooled exports (five causes, before and after 2020) and 2 more for the exploratory Alzheimer's county run, plus 4 year-by-year exports kept for the diabetes and overdose comparison and 2 heart disease 2018-2019 exports showing the two databases match exactly county by county. No synthetic data is used anywhere. The full method, the predictions made before seeing the data, and every change made along the way are in research_protocol.md. Export steps are in manual_data_acquisition.md. The interactive app and source code are at github.com/aidanc667/covid-mortality-disruption-lab.

This project uses public aggregate data for research and education. It isn't medical advice and doesn't predict risk for any individual.